

Shashidhar T

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PROFESSIONAL SUMMARY:

Full Stack Developer with **5+** years of experience designing and delivering scalable, secure, and high-performance applications across finance, healthcare, and enterprise domains. Proficient in **Python (Django, Flask, FastAPI), Node.js, React, Angular, and TypeScript**, with hands-on expertise in **AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Redis, Kafka, and PostgreSQL/Oracle/MongoDB**. Skilled in **REST/GraphQL APIs, microservices, event-driven systems, and automated testing**, delivering measurable improvements in system performance, deployment efficiency, and reliability. Proven record of reducing deployment time by **40–70%**, improving system throughput by **30%**, and delivering highly reliable production systems. Currently exploring the integration of **AI/ML** models into full-stack applications for intelligent automation and predictive analytics. Passionate about clean code, agile collaboration, and solving complex technical challenges at scale.

TECHNOLOGIES:

Programming & Web:	Python 3.9/3.10, Node.js, JavaScript, TypeScript, HTML5, CSS3, React.js 18-16, Next.js, Angular 16-8
Frameworks & Libraries:	Django, Flask, FastAPI, Node.js, Pandas, NumPy, Matplotlib, SQLAlchemy, Django ORM, PySpark.
Cloud & DevOps:	AWS (EC2, S3, RDS, Lambda ,EKS, IAM), Azure(AKS), GCP, Docker, Kubernetes, Terraform, CI/CD pipelines (Git/GitLab/Jenkins)
Databases & Caching:	PostgreSQL, MySQL, MongoDB, Oracle, DynamoDB, Redis, Elasticsearch.
Event-Driven / Messaging Systems:	Kafka, RabbitMQ.
Security & Authentication:	OAuth 2.0, JWT, Role-Based Access Control (RBAC), HIPAA Compliance.
Testing & QA:	Pytest, unittest, Selenium, Postman, TDD
Monitoring & Logging:	Datadog, Prometheus, Splunk, ELK Stack (Elasticsearch, Logstash, Kibana)
Project Management & Collaboration:	Jira, Confluence, Slack, Agile Scrum
Software Engineering & Methodologies:	Microservices Architecture, REST APIs, GraphQL, MVC, CI/CD, Problem-Solving, Data Structures, Algorithm Analysis

PROFESSIONAL EXPERIENCE:

Python Full Stack Developer
ResMed Plano, TX

Apr 2025 - Present

- Designed and developed FHIR-compliant **REST APIs** using **Python, Django**, and **FastAPI** to securely process and exchange over **200K+ patient healthcare records daily**, supporting real-time clinical workflows and analytics platforms.
- Built scalable backend services following **microservices architecture**, designing business logic, data models, and service communication layers.
- Integrated **OAuth 2.0** and **JWT** authentication mechanisms to enforce **HIPAA-compliant security**, role-based access control, and secure API communication across services.
- Developed modern **React18** and **TypeScript** frontend applications with responsive **UI** using **HTML5** and **CSS3**, delivering real-time clinical dashboards and workflows.
- Integrated frontend applications with backend APIs using **JSON**-based communication, enabling seamless data exchange across systems.
- Designed and optimized **PostgreSQL** and **Oracle** databases to manage transactional data and large-scale datasets efficiently.
- Implemented event-driven processing using **Kafka**, enabling real-time system updates and reliable message handling between microservices.
- Created data visualizations using **Matplotlib** to generate clinical trend reports and performance metrics for healthcare stakeholders.
- Deployed and managed applications on **AWS** cloud services including **EC2, S3, RDS, Lambda, CloudFront, EKS, Route53, IAM**, and Secrets Manager, ensuring scalability and secure infrastructure.
- Implemented **Redis** for caching frequently accessed data, improving application performance and reducing backend load.
- Automated routine deployment and maintenance tasks using **Bash/Shell** scripts on **AWS EC2** instances, improving operational efficiency and reducing manual effort.
- Automated infrastructure setup and deployments using **Terraform (Infrastructure as Code)** and **CI/CD** pipelines to streamline releases.
- Conducted comprehensive backend testing using **Pytest** and **unit test**, performing API, integration, and workflow testing for **FHIR** and **HL7** data pipelines, maintaining **90%+ test coverage**.
- Monitored application performance and system health using **Datadog, Splunk**, and **ELK stack**, proactively identifying issues and reducing incident response times by **40%**.
- Monitored application health and performance using enterprise monitoring tools, proactively identifying and resolving production issues.
- Collaborated in **Agile Scrum teams** using **Jira, Slack**, and **Git**, contributing to sprint planning, code reviews, and cross-team delivery.

Environment: Python 3.10, Django, FastAPI, REST APIs, Microservices, OAuth2/JWT, Kafka, Matplotlib, React18, TypeScript, HTML5, CSS3, JSON, PostgreSQL, Oracle, Redis, AWS (EC2, S3, RDS, Lambda, CloudFront, EKS, Route53, IAM, Secrets Manager), Docker, Terraform, CI/CD pipelines, Bash/Shell scripts, Datadog, Splunk, ELK Stack, Git/GitLab, Agile Scrum, Jira, Slack, Pytest, unittest

- Developed **Python**-based microservices using **Django/Flask**, and **FastAPI** to process and manage over 1M+ daily banking transactions, ensuring high availability and low latency.
- Built **RESTful APIs** integrated with **Angular16** dashboards (**JavaScript, HTML5, CSS3**) for real-time transaction monitoring, account management, and analytics, improving internal reporting efficiency by **35%**.
- Implemented **OAuth2/JWT**-based authentication with role-based access control to secure sensitive banking data, ensuring 100% compliance with regulatory security standards.
- Designed and optimized **Oracle** and **MongoDB** databases using **SQLAlchemy ORM**, reducing query execution time by **30%** for high-volume financial datasets.
- Developed **event-driven workflows** using **GCP Pub/Sub** and **Kafka**, enabling real-time transaction updates across multiple banking systems and reducing data latency by **25–30%**.
- Built **ETL** pipelines using **PySpark** and **Python** to migrate legacy transaction data into modern analytics systems, achieving **99% data accuracy** and supporting real-time reporting.
- Containerized backend services with **Docker** and deployed on **GCP Kubernetes Engine (GKE)**, achieving **99.9% uptime** for critical banking services.
- Implemented **Redis caching** for frequently accessed transaction data, improving API response times by **40%** and reducing backend load.
- Automated deployments and infrastructure provisioning using **Terraform** and **CI/CD pipelines**, reducing manual deployment time by **60%**.
- Monitored system health and application performance using **Prometheus**, and **Splunk**, proactively identifying and resolving bottlenecks, reducing incident response times by **35%**.
- Collaborated in **Agile Scrum teams** using **Jira** and **Confluence**, conducted code reviews, mentored junior developers, and maintained API and system workflow documentation, improving team efficiency and knowledge sharing by 20%.
- Created data visualizations using **Matplotlib** to display transaction trends, fraud alerts, and performance metrics, improving actionable insights for business stakeholders.

Environment: Python 3.9/3.10, Django, Flask, FastAPI, REST APIs, Microservices, MVC, OAuth2/JWT, SQLAlchemy, GraphQL, Angular16, TypeScript, HTML5, CSS3, JSON, Oracle, MongoDB, GCP (Compute Engine, Cloud SQL, Cloud Storage, Pub/Sub, Kubernetes Engine), Docker, Kubernetes, CI/CD (Git, Jenkins, Terraform, PySpark, Python ETL scripts, Prometheus, Splunk, Jira, Confluence, Pytest, unittest.

- Designed and developed a real-time campus service workflow management web application used by 1,200+ students and staff for IT, housing, and administrative service requests.
- Built backend microservices using **Python (Flask/FastAPI)** to process requests, authentication, and notifications, handling 3,000+ transactions per week.
- Developed responsive frontend interfaces using **Next.js (React)** and **JavaScript**, improving user engagement and request submission efficiency by 35%.
- Enhanced **UI/UX** components and optimized page rendering using **Next.js** features, improving usability and responsiveness.
- Integrated **RESTful APIs** and **AJAX** for real-time updates across system modules, reducing request status update latency by 40%.
- Deployed and maintained the application on **AWS (EC2, S3, RDS)** with **PostgreSQL/MySQL**, supporting high availability and secure data storage.
- Implemented **JWT**-based role access control, reducing unauthorized access incidents by 100%.
- Used **Redis** caching and background processing to optimize performance, decreasing API response times by 30%.
- Containerized services using **Docker**, ensuring consistent development and production environments.
- Performed automated API testing with **Pytest** and **Postman**, and automated frontend **UI** testing with **Selenium**, improving system stability and reducing defects by **25%**.
- Managed code versioning using **Git** and participated in **agile sprint** cycles.
- Monitored application performance and system health, achieving 99.5% uptime during active semesters.

Environment: Python 3.9/3.10, Flask, FastAPI, REST APIs, Microservices, JWT Authentication, Next.js (React), JavaScript, HTML5, CSS3, AJAX, PostgreSQL, MySQL, Redis (caching), AWS (EC2, S3, RDS), Docker, CI/CD pipelines, Pytest, Postman, Selenium, Git, Agile (Jira).

EDUCATION:

Master of Science in Data Science | Lewis University, Romeoville, IL.